SU
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INT 453: Digital Multimedia

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Lecture 6

Introduction to Transformation Techniques

1- DCT

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books

- Fundamentals of Multimedia.

Overview

This Lecture covers:

- Why Digital Transformation
- Frequency Transforms Functionality
- Wavelet Transform Functionality
- DCT concept
- DCT Examples

Introduction (cont.,)

Lossless compression algorithms do not deliver *compression ratios* that are high enough.

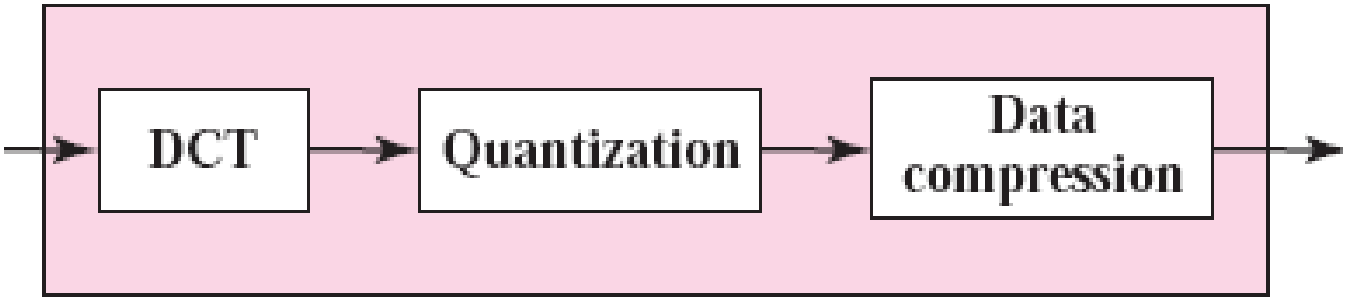
Hence, most multimedia compression algorithms are *lossy*.

What is *lossy compression*?

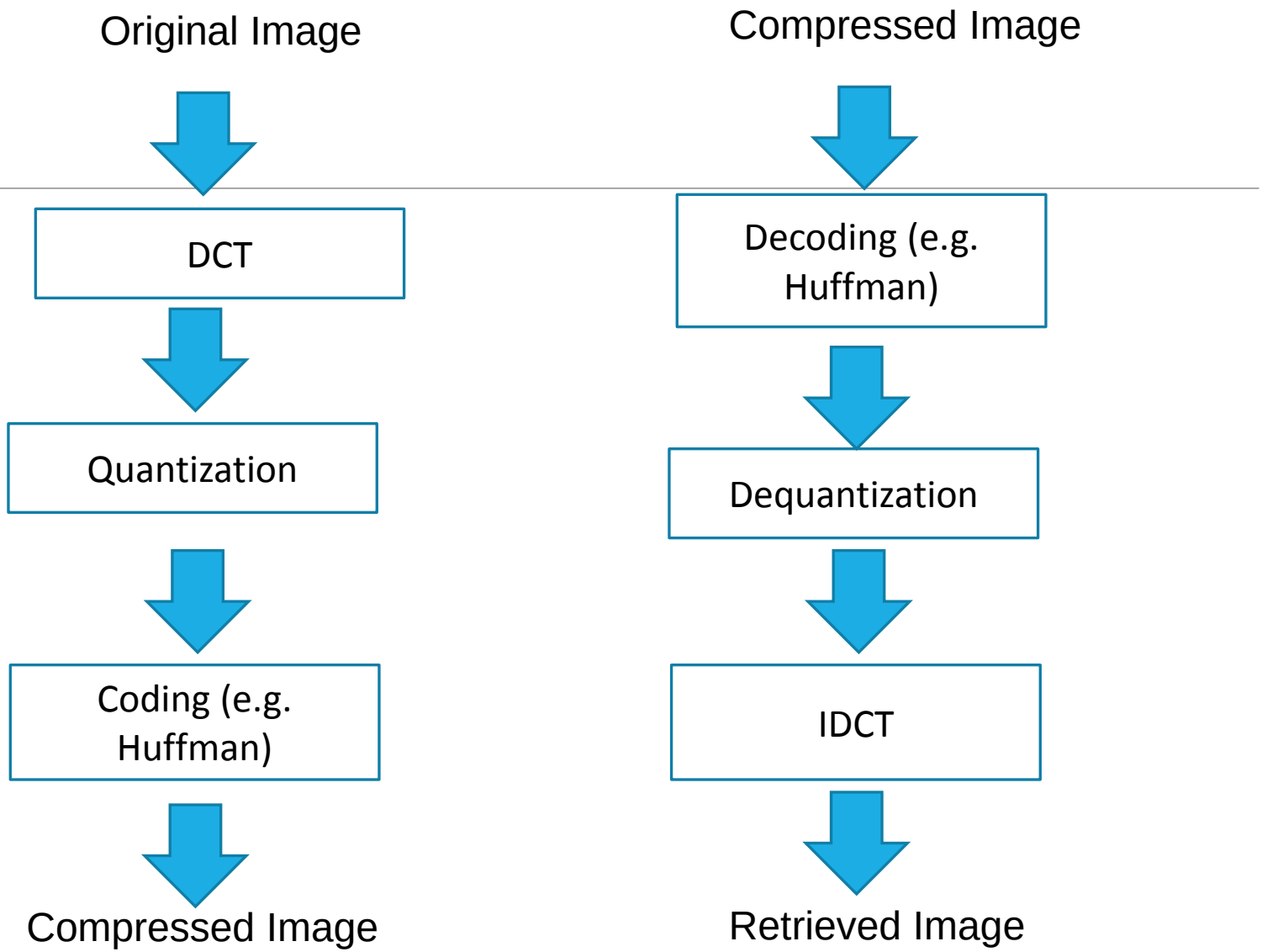
- The compressed data is not the same as the original data, but a close approximation of it.
- Yields a much higher compression ratio than that of lossless compression.

Lossy Compression Phases

- Transform Coding (i.e., DCT)
- Quantization
- Data Compression



Transform Coding (i.e., DCT)



In **lossy compression**, the retrieved image is not the original image

Transform Coding (DCT)

- It is the step number one in lossy compression
  - From basic principles of information theory, we know that coding vectors is more efficient than coding scalars.
  - To carry out such an intention, we need to group blocks of consecutive samples from the source input into vectors.

Transform Coding (DCT)

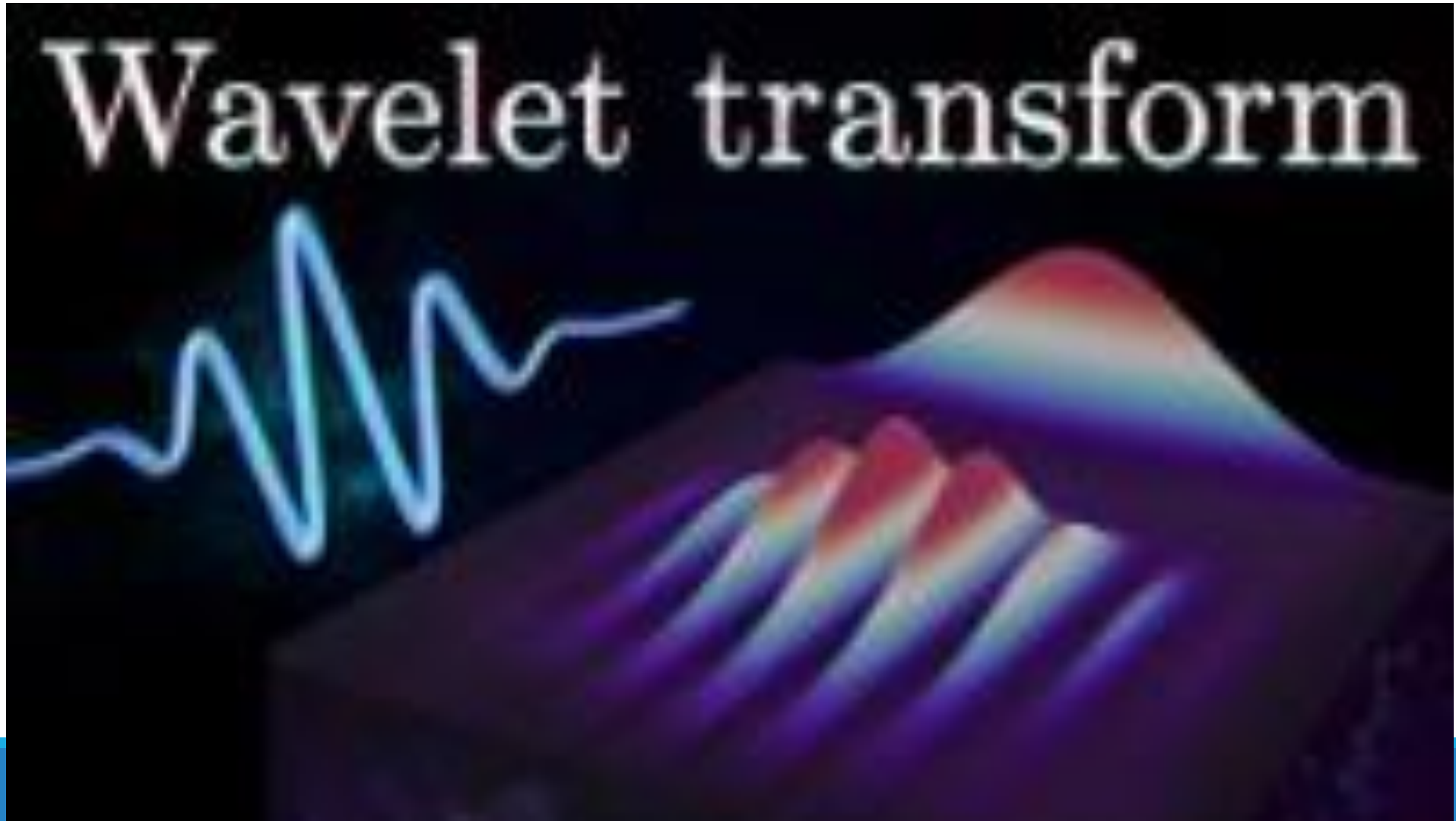
- Why transformation is important?
  - If **Y** is the result of a linear transform **T** of the input vector **X** in such a way that the components of **Y** are much less correlated, then **Y** can be coded more efficiently than **X**.
  - If most information is accurately described by the first few components of a transformed vector,
  - then the remaining components can be quantized (set to zero), with little signal distortion.

Why Transform Coding?

This lecture will be discussed through the following video.

https://www.youtube.com/watch?v=jnxqHcObNK4

Why Transform Coding?



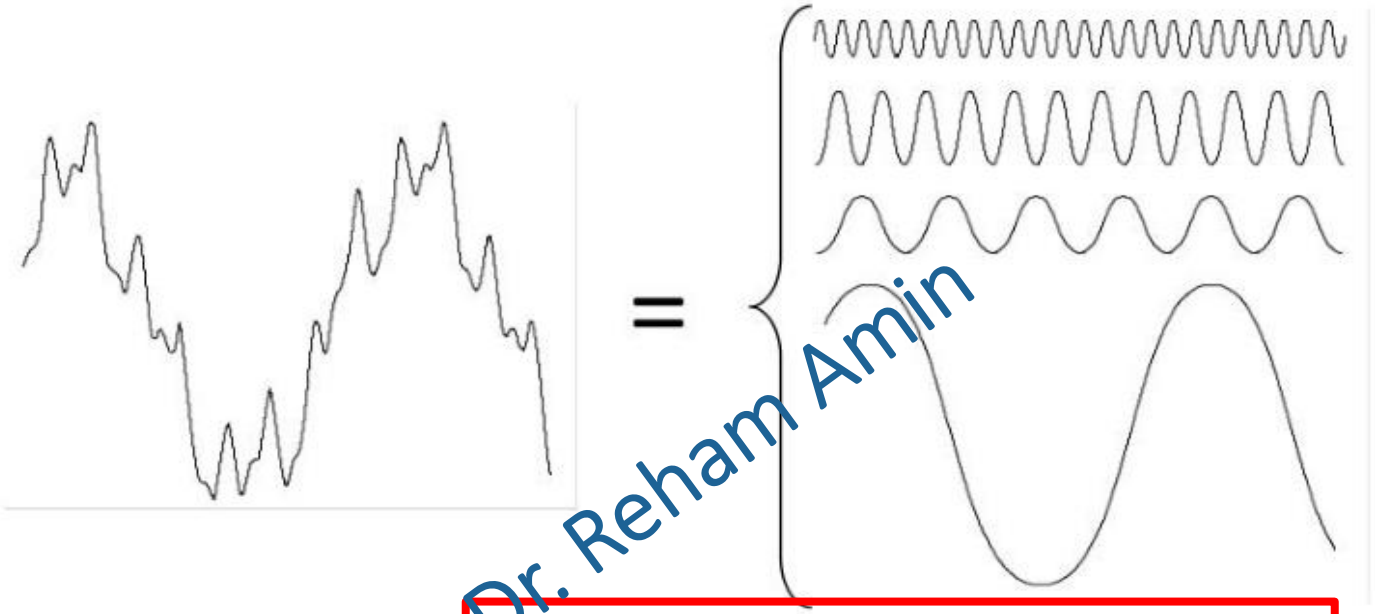
Discrete Cosine Transform (DCT)

- DCT is a widely used as transform coding technique
- It is able **to perform decorrelation of the input signal in a data-independent manner** (to facilitate quantization).
- Given an input function $f(i, j)$ over two integer variables i and j (a piece of an image, for example 8x8),
- The 2D DCT transforms it into a new function $F(u, v)$, with integer u and v running over the same range as i and j.

Discrete Cosine Transform (DCT)

- The discrete cosine transform (DCT) represents an image as sum of sinusoids of varying magnitudes and frequencies.
- DCT is a variant of DFT
- DFT having complex values, while DCT have real values instead.
- DCT is equivalent of computing 2N-point DFT of a signal which is made even symmetric by its extension.
- DCT coefficients are real valued while DFT coefficients are complex, therefor implementation of DCT is easier than DFT.

The Big Idea



Any function that periodically repeats itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient – a *Fourier series*

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Mathematical background

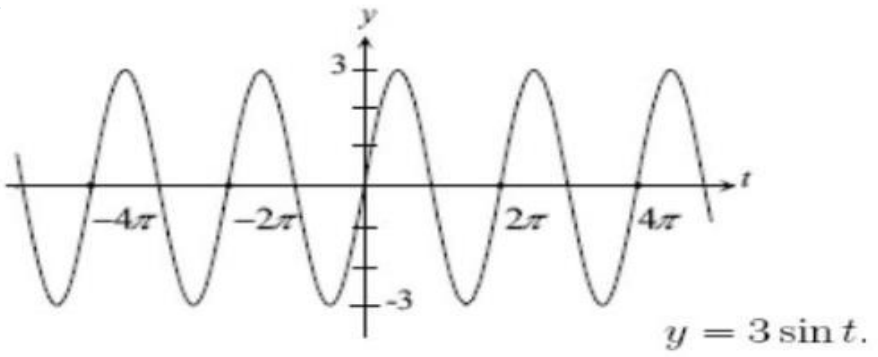
Sine and Cosine Functions

- Periodic functions
- General form of sine and cosine functions:

$$y(t) = A\sin(\alpha t + b) \qquad y(t) = A\cos(\alpha t + b)$$

$\lvert A\rvert$	amplitude
$\frac{2\pi}{\lvert\alpha\rvert}$	period
b	phase shift

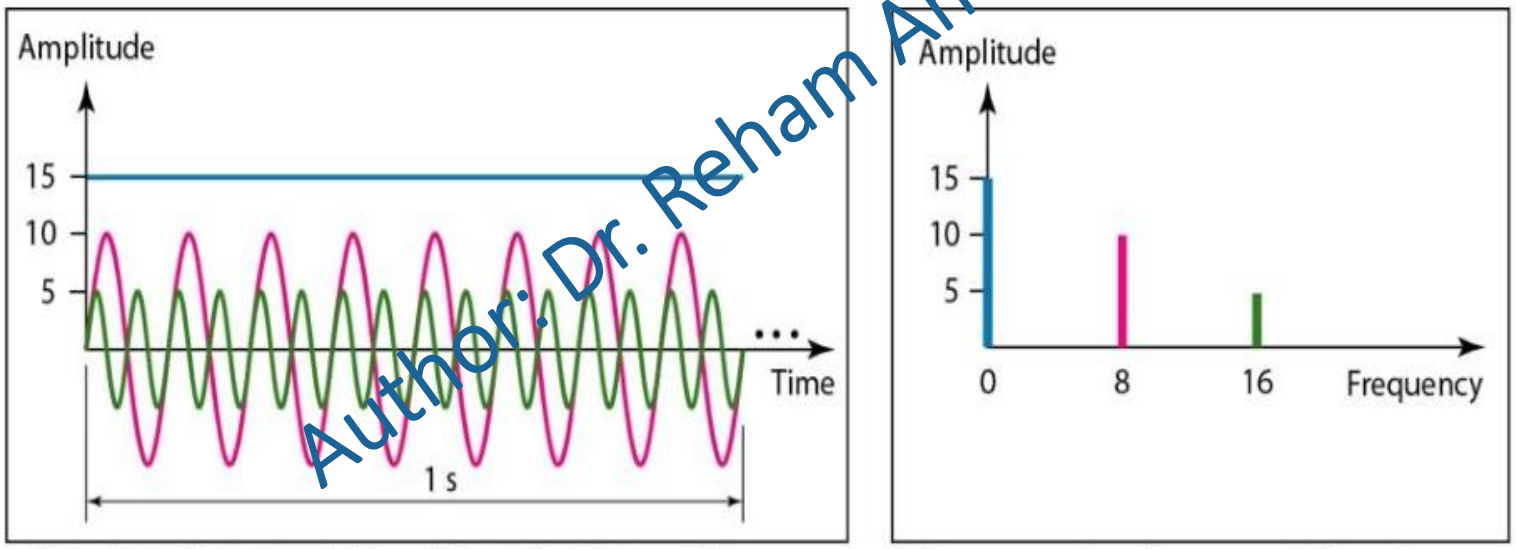
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$y = 3\sin t.$

Mathematical background

The time domain and frequency domain of three sine waves



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16

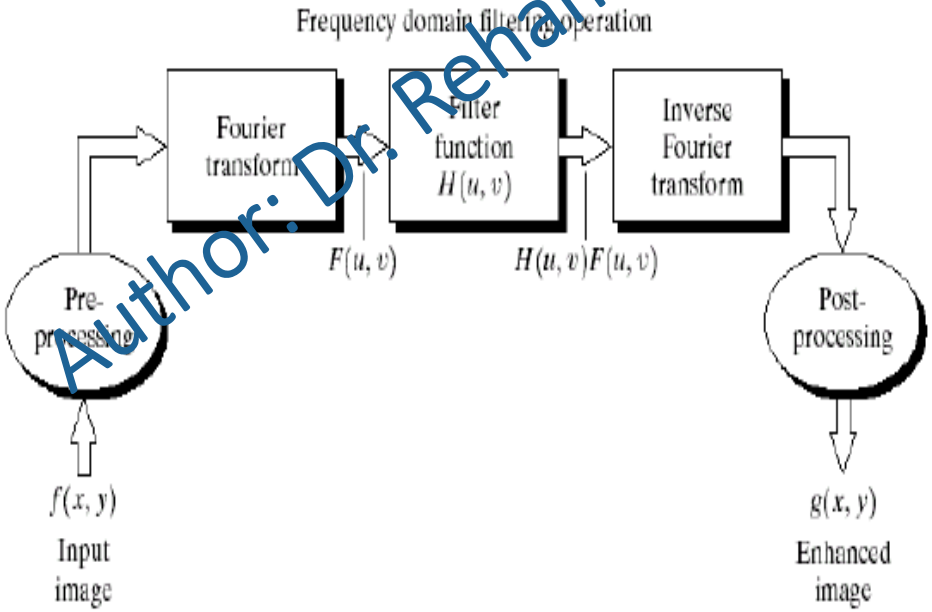
b. Frequency-domain representation of the same three signals

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The DFT and Image Processing

To filter an image in the frequency domain:

1. Compute $F(u,v)$ the DFT of the image
2. Multiply $F(u,v)$ by a filter function $H(u,v)$
3. Compute the inverse DFT of the result



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Image processing in the frequency domain

The image is firstly transformed using discrete Fourier transform (which can be computed efficiently using fast Fourier transform (FFT)).

The Fourier coefficients are then adjusted according to different imaging tasks.

The restored image can be obtained by taking the inverse Fourier transform (which can again be computed efficiently using FFT).

In fact, instead of using DFT, wavelet transform (such as Haar transform) can also be applied.

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Discrete Cosine Transform (DCT)

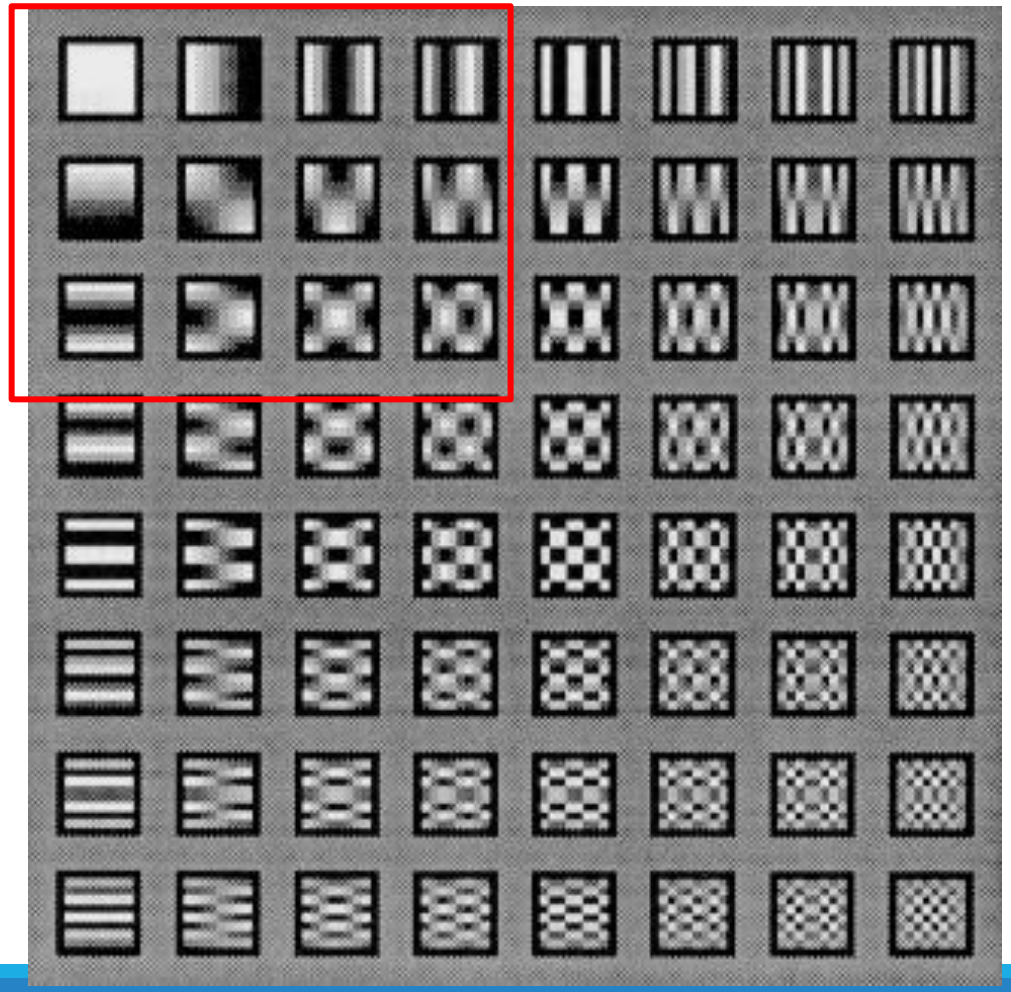
- Also FFT structure can be used to make DCT computation faster, therefore it has become choice for many applications.
- The DCT has the property that, for a typical image**, most of the visually significant information about the image is concentrated in just a few coefficients of the DCT**.
- This property **is known as Energy Compaction property**. Therefore, DCT finds it's importance in image and video compression techniques.
- DCT is a heart of the international standard lossy image compression algorithm known as JPEG.
- DCT has vast application areas, such as:
- Image compression: JPEG, HEIF, WebP, BPG etc.
- Video compression : H.261, MJPEG, MPEG1, H.263, H. 264 (AVC MPEG) , H.265 (HEVC), WebM etc.
- Audio Compression : Ac3 (Dolby), AC4, MP3, MP$, AAC, AAC+, WMA etc.
- Speech Coding (VOIP), Digital Radio etc.

Discrete Cosine Transform (DCT)
The basic operation of the DCT

- For most images, much of the signal energy lies at low frequencies; these appear in the **upper left corner** of the DCT.
- Eye less sensitive to high frequencies and more at low frequencies.
- Compression is achieved since the lower right values **represent higher frequencies**, and are often small - small enough to be neglected with little visible distortion.
- The DCT input is an (8 by 8 array) of integers.
- This array contains each pixel's gray scale level; 8 bit pixels have levels from 0 to 255.

Discrete Cosine Transform (DCT)

much of the signal energy lies at low frequencies



Discrete Cosine Transform (DCT)

- A common issue with DCT in digital media are blocky compression, caused by DCT blocks.
- Blocky compression artifacts can appear when heavy DCT compression is applied.
- Truncation of higher spectral coefficients results in bluring of the images, especially wherever the details are high.
- Corse quantization of some of the low spectral coefficients introduces grannies in the smooth portions of the images.

Discrete Cosine Transform (DCT)

The general definition of the transform is:

$$F(u,v) = \frac{2\,C(u)\,C(v)}{\sqrt{MN}} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} \cos\frac{(2i+1)\cdot u\pi}{2M} \cdot \cos\frac{(2j+1)\cdot v\pi}{2N} \cdot f(i,j)$$

where $i, u = 0, 1, \ldots, M-1$; $j, v = 0, 1, \ldots, N-1$; and the constants $C(u)$ and $C(v)$ are determined by

$$C(\xi) = \begin{cases} \frac{\sqrt{2}}{2} & if \quad \xi = 0, \\ 1 & otherwise. \end{cases}$$

F(u, v) is a transformed pixel, you need **four** loops to determine the transformed image

2D Discrete Cosine Transform (2D DCT)

$$F(u,v) = \frac{C(u)\,C(v)}{4}\sum_{i=0}^{7}\sum_{j=0}^{7}\cos\frac{(2i+1)u\pi}{16}\cos\frac{(2j+1)v\pi}{16}f(i,j)$$

where $i, j, u, v = 0, 1, \ldots, 7$, and the constants $C(u)$ and $C(v)$ are determined by the previous equation.

2D Inverse Discrete Cosine Transform (2D IDCT):

The inverse function is almost the same, with the roles of $f(i, j)$ and $F(u, v)$ reversed, except that now $C(u)C(v)$ must stand inside the sums

$$\tilde{f}(i,j) = \sum_{u=0}^{7}\sum_{v=0}^{7}\frac{C(u)\,C(v)}{4}\cos\frac{(2i+1)u\pi}{16}\cos\frac{(2j+1)v\pi}{16}F(u,v)$$

where $i, j, u, v = 0, 1, \ldots, 7$.

Optimized method to find 2D DCT

$$F(u, v) = t(u, i)\ \mathbf{f(x, y)}\ t^{T}(u, i)$$

$$t(u,i) = \begin{cases} \frac{1}{\sqrt{n}} & u = 0 \\ \sqrt{\frac{2}{n}} \cos\left(\frac{(2i+1)u\pi}{2n}\right) & u \neq 0 \end{cases}$$

Example 1

$$t(u,i) = \begin{cases} \frac{1}{\sqrt{n}} & u = 0 \\ \sqrt{\frac{2}{n}} \cos\left(\frac{(2i+1)u\pi}{2n}\right) & u \neq 0 \end{cases}$$

Find DCT for the following block 4x4 of a gray scale image

	136	142	148	161
f(x, y)=	138	145	149	164
	131	143	150	158
Solution	135	139	149	162

F(u, v) = t(u, i) **f(x, y)** t^T(u, i)

Image is 4*4 so m=n=4

	i=0	i=1	i=2	i=3
u = 0	0.5	0.5	0.5	0.5
u = 1	0.653281482	0.27059805	-0.27059805	-0.653281482
u = 2	0.5	-0.5	-0.5	0.5
u = 3	0.27059805	-0.6532815	0.653281482	-0.27059805

Example 1 (Cont'd)

Use DCT Excel file

F(u, v) =

587.50	-37.95	5.00	-5.39
2.55	1.96	0.97	-1.81
-1.50	-0.35	3.00	1.77
-4.30	-0.31	-4.19	0.54

F(u, v) =

588	-38	5	-5
3	2	1	-2
-1	0	3	2
-4	0	-4	1

F(u, v) is jointly correlated u and v

Example 2

$$t(u,i) = \begin{cases} \frac{1}{\sqrt{n}} & u = 0 \\ \sqrt{\frac{2}{n}} \cos\left(\frac{(2i+1)u\pi}{2n}\right) & u \neq 0 \end{cases}$$

Find DCT for the following block 3x3 of a gray scale image (similar to DC signal)

f(x, y)=

20	20	20
20	20	20
20	20	20

Solution

$F(u, v) = t(u, i)\, f(x, y)\, t^T(u, i)$

n=3

	i=0	i=1	i=2
u=0	0.577350269	0.57735027	0.577350269
u=1	0.707106781	0	-0.707106781
u=2	0.40824829	-0.8164966	0.40824829

Example 2

$$t(u,i) = \begin{cases} \frac{1}{\sqrt{n}} & u = 0 \\ \sqrt{\frac{2}{n}} \cos\left(\frac{(2i+1)u\pi}{2n}\right) & u \neq 0 \end{cases}$$

f(x, y)=

20	20	20
20	20	20
20	20	20

Solution

t(u, i)

	i=0	i=1	i=2
u=0	0.577350269	0.57735027	0.577350269
u=1	0.707106781	0	-0.707106781
u=2	0.40824829	-0.8164966	0.40824829

	i=0	i=1	i=2
u=0	1	1	1
u=1	1	0	-1
u=2	0	-1	0

Example 2

	i=0	i=1	i=2
u=0	1	1	1
u=1	1	0	-1
u=2	0	-1	0

20	20	20
20	20	20
20	20	20

t(u, i) f(x, y)

60	60	60
0	0	0
-20	-20	-20

Example 2

$t(u, i)\ f(x, y)\ t^T(u, i)$

60	60	60
20	20	20
0	0	0

1	1	0
1	0	-1
1	-1	0

180	0	-60
60	0	-20
0	0	0

Review Questions

1. Explain in detail the benefits of using the transformation techniques.
2. DCT has vast application areas mention 3 of these applications and explain the benefits of it.

Review Questions (1)

Find DCT for the following block 3x3 of a gray scale image

f(x, y)=

20	30	40
20	30	40
20	30	40

Review Questions (2)

Find DCT for the following block 8x8 of a gray scale image

f(x, y) =

200	202	189	188	189	175	175	175
200	203	198	188	189	182	178	175
203	200	200	195	200	187	185	175
200	200	200	200	197	187	187	187
200	205	200	200	195	188	187	175
200	200	200	200	200	190	187	175
205	200	199	200	191	187	187	175
210	200	200	200	188	185	187	186

Solution

$F(u, v) = t(u, i)\, f(x, y)\, t^{T}(u, i)$

Assignment

Answer the following questions

1. Why are signals (audio/ image) transformed from time domain to frequency domain?
2. Why wavelet is preferred over frequency domain transforms?
3. How wavelet transform work?

Assignment

Answer the following questions

1. What are the disadvantages of DCT?
2. What are blocky compression artifacts?
3. What is image blurring, and why it happened?
4. What is the sinusoid function?
5. Solve all the mathematical exercises in the lecture besides the review questions, and specify the time that is required for each one.

Questions?

Thank You